Ashtyn DeNovellis

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Greeley, Colorado

Education

University of Northern Colorado — Greeley, Colorado
Master of Science in Biological Sciences — *Expected May 2028*

- **Relevant Coursework:** Cell Biology, Biochemistry, Genetics, Microbiology.

University of Northern Colorado — Greeley, Colorado
Bachelor of Science in Software Engineering, Summa Cum Laude (GPA: 3.97) — *May 2026*

- **Relevant Coursework:** AI, Data Structures & Algorithms, Database Management, UI/UX Design, Project Management, PHP Development, Web Design, Object Oriented Programming, Operating Systems.
- **Honors/Awards:** UNC Presidential Trustee Scholarship, Dean's List every semester, Academic Excellence Awards.

Projects

Mexican Gray Wolf Genetic Diversity | *Python, Pandas, NumPy, Matplotlib* — October 2026

- Processed PLINK pedigree datasets (MW.clean.ped) using Pandas to calculate individual genetic heterozygosity rates across populations.
- Generated a pairwise breeding recommendation matrix heatmap to support animal sanctuary conservation and wildlife breeding programs.

Waterfowl Local BLAST Database | *BLAST+, SRA Toolkit* — October 2026

- Compiled a 15-million sequence local BLAST database from a 2.43GB SRA waterfowl dataset (SRR38913815).
- Executed tblastn searches utilizing bacteriophage major capsid protein bait sequences to successfully isolate and identify 500 biological alignment hits.

Microbiome FASTQ Sequence Pipeline | *Python, Biopython, Matplotlib* — September 2026

- Developed a memory-efficient sequence analysis script utilizing Biopython's FastqGeneralIterator to process 2.89M whole-genome sequencing reads.
- Calculated average GC fractions and plotted distribution histograms to successfully identify and map secondary contamination peaks ($>$0.73 GC).

Eco-Dex | *React, Firebase, Gemini AI, Vercel* — August 2025 – May 2026

- Building a full-stack application to process biological data, identifying 10,000+ species with real-time conservation status.
- Incorporated Gemini AI API for image-based identification, returning biological classifications in under 15 seconds.
- Created a responsive React UI from 15+ Figma screens and implemented Firebase for secure user data management.

Bear Events | *React, Agile (Scrum), Jira* — August 2025 – October 2025

- Architected a full-stack platform featuring 15+ reusable React components and real-time tracking for 12+ event types.
- Collaborated in a 30-person Agile environment using Jira and Git, conducting extensive code reviews to merge features.

AI Isolation Game Algorithm | *C++, Monte-Carlo Algorithm* — January 2025 – May 2025

- Created a custom Monte-Carlo algorithm for an AI bot, achieving a 90% win rate against baseline models.
- Optimized heavy C++ implementation for strict memory efficiency and low decision latency in tournament environments.

Guinea Games | *Python, Pygame, SQLite* — October 2025 – December 2025

- Led a 16-person front-end team for a 2D application, engineering core logic and computational collision detection.
- Produced 15+ 2D art assets and integrated state with SQLite to securely store and retrieve user progress datasets.

AI Powered Language Learning | *React, MongoDB, Ollama API* — January 2025 – May 2025

- Engineered a React-based web application with a real-time asynchronous chat interface and sub-10s response times.
- Blended Ollama AI RESTful APIs for instant feedback and built a MongoDB database for conversation history.

Technical Skills

Languages: Python, C++, SQL, Java, C#, JavaScript, TypeScript
Frameworks/Libraries: .NET, React, Node.js, Pandas, NumPy, Biopython, Matplotlib
APIs/Databases: Firebase, MongoDB, SQLite, Gemini API, Ollama API
Tools: GitHub, Visual Studio, VS Code, BLAST+, SRA Toolkit, Figma, Blender, Jira, Microsoft Teams
Algorithms: Simulated Annealing, Breadth/Depth First Search, Monte Carlo, Ant Colony
Methodologies: Agile Development (Scrum, Kanban)